

Domestic Bot Dashboard

(Assistive Robot OS Flow)

Submitted By

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MINI LAB PROJECT REPORT

This Report Presented in Partial Fulfillment of the course **CSE441: UI and UX Design** in the Computer Science and Engineering Department



DAFFODIL INTERNATIONAL UNIVERSITY
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DECLARATION

I hereby declare that this lab project has been done by me under the supervision of **Mr. Md. Hasanuzzaman Dipu, Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. I also declare that neither this project nor any part of this project has been submitted elsewhere as a lab project.

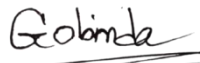
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COURSE & PROGRAM OUTCOME

The following course has course outcomes (CO's) as stated below:

Table 1: Course Outcome Statements

CO's	Statements
CO1	Define and relate screens, components, and state relationships needed for solving a specific interaction-design problem.
CO2	Formulate knowledge of front-end development and UI/UX design principles in problem solving.
CO3	Analyze user research, wireframes, and interface models to present a specific problem.
CO4	Develop solutions for real-world complex problems applying front-end and UI/UX concepts while evaluating their effectiveness based on industry standards.

Table 2: Mapping of CO, PO, Blooms, KP and CEP

CO	PO	Blooms	KP	CEP
CO1	PO1	C1, C2	KP3	EP1, EP3
CO2	PO2	C2	KP3	EP1, EP3
CO3	PO3	C4, A1	KP3	EP1, EP2
CO4	PO3	C3, C6, A3, P3	KP4	EP1, EP3

The mapping justification of this table is provided in sections 4.3.1, 4.3.2 and 4.3.3.

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Chapter 1

Introduction

This chapter introduces the Domestic Bot Dashboard project, describing its background, motivation, objectives, feasibility, the gap it addresses, and its expected outcome.

1.1 Introduction

Domestic Bot Dashboard is an interactive, touch-based museum kiosk exhibit that lets visitors control a simulated assistive household robot and watch it complete everyday domestic tasks step by step. Museum visitors typically have only a short attention span and no prior robotics knowledge, so most engineering-style robot control panels are too dense and technical for them to use unassisted. The system solves this by presenting a friendly, icon-driven dashboard where a visitor picks a task — cleaning a room, delivering an item, or sending the robot to a location — configures it in one or two taps, and watches the robot execute the task with live progress feedback, all within roughly 90 seconds.

1.2 Motivation

Understanding how a household robot plans and executes a task is difficult to convey through static posters or text panels, because the process normally happens out of sight. This project was motivated by the idea of making that process visible and interactive: by letting visitors choose a task themselves, configure it, and watch a live progress bar and status messages as the simulated robot works, the underlying concept of human-robot collaboration becomes far easier to grasp. The project also provided an opportunity to apply UI/UX design and front-end web development skills to a genuine science-communication and accessibility problem.

1.3 Objectives

- To design an accessible, touch-first kiosk dashboard that a first-time visitor can understand within approximately 90 seconds.
- To let visitors choose from a small set of realistic domestic robot tasks (clean a room, deliver an item, go to a location) and configure each one in a couple of taps.
- To show a clear, live task-execution view with a progress bar and plain-language status updates.
- To provide a dedicated, always-reachable Emergency Stop control for visitor safety.
- To support multiple languages and ADA-aligned accessibility settings (high contrast, large text, voice guidance, kiosk reach mode).
- To apply UI/UX principles — visual hierarchy, Fitts' Law, feedback, and consistency — to a realistic museum exhibit interface.

1.4 Feasibility Study

Interactive kiosks are widely used in science museums to let visitors learn through direct manipulation rather than passive reading, and consumer robot apps (robot vacuum companions, smart-home hubs) already prove that a small set of clear status indicators and task buttons is enough for non-technical users to operate a robot confidently. On the technical side, a modern browser-based front-end stack (React 19 with TypeScript, Tailwind CSS, and the Motion animation library) makes it feasible to build a fully self-contained, offline-capable kiosk application, with the robot's task execution simulated in the browser rather than requiring physical robot hardware for the exhibit prototype. This was confirmed by building and running a working prototype of the interface over the course of the project, including deploying it as a live web build.

1.5 Gap Analysis

Most existing robot-control interfaces fall into two extremes: dense engineering dashboards built for developers, or oversimplified toy-like apps that hide how the robot is actually completing a task. Public museum exhibits about robotics, in turn, rarely let a visitor directly configure and dispatch a simulated robot and watch clear, live feedback on its progress. Domestic Bot Dashboard addresses this gap by combining a simple, task-driven touch interface with a two-step configuration flow and a live progress view, wrapped in full multilingual and accessibility support, in a single self-contained kiosk application.

1.6 Project Outcome

The expected outcome of the project is a functional, user-friendly kiosk web application that lets a visitor select a domestic task, configure it (choosing a room or item), confirm it, and watch the robot execute it with live progress feedback and a clear completion summary. The project demonstrates how interactive visualization and user-centred interface design can make an otherwise abstract topic — autonomous robot task execution — engaging and understandable for a general museum audience within a very short interaction window.

Chapter 2

Research & User Journey (UX)

This chapter presents the target audience, representative user personas, the visitor journey map, the overall user flow, and the accessibility plan behind the Domestic Bot Dashboard exhibit.

2.1 Target Audience

The primary audience for Domestic Bot Dashboard is science-museum visitors of all ages interested in robotics and artificial intelligence, including children, general adults, elderly visitors, and technically curious students. A large touch-friendly interface, plain-language task cards, and clear iconography allow visitors of different ages and technical comfort levels to explore the exhibit independently, without staff assistance or prior instructions.

2.2 User Personas

Persona	Goal	Tech Familiarity	Main Pain Point
Rima Hasan, 34 General Public	Try the robot quickly and share the moment with her child	Medium (smartphones daily)	No patience for instructions or small text
Adnan, 9 Child Visitor	Press buttons, see the robot move and finish a task	High (touch), low (reading)	Gets bored fast; can't read long text
Mrs. Karim, 68 Elderly Visitor	Understand what the robot does out of curiosity	Low	Small text/buttons hard to see and tap
Farhan, 22 Engineering Student	Understand how task configuration and execution work	High	Wants more detail than casual visitors see

2.3 User Journey Map

Table 3: Visitor Journey Map

Stage	Visitor Action	Visitor Experience
Discover	Notices the Domestic Bot kiosk and its friendly robot avatar	Becomes curious and approaches the screen
Approach	Reads the title and taps “Start Exploring”	Understands this is an interactive, not passive, exhibit
Orient	Sees the Main Dashboard: robot status, battery, AI suggestion	Feels oriented and in control of the exhibit
Select	Taps a Quick Action or “View All Tasks” to choose a task type	Feels in control of what the robot will do

Stage	Visitor Action	Visitor Experience
Configure	Picks a room or item and reviews the task summary	Understands exactly what will happen before confirming
Observe	Watches the live progress bar and status messages	Sees the robot's work become visible and understandable
Reflect	Reviews the task execution summary and science takeaway	Understands what just happened and why it matters
Finish	Selects another task or returns to the dashboard	Leaves with a clear, positive impression of assistive robotics

2.4 User Flow Diagram

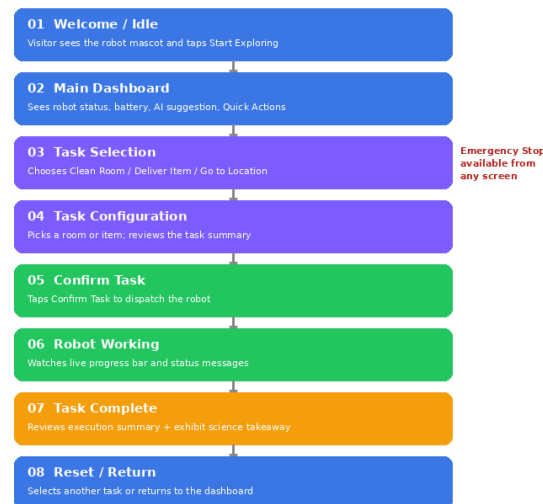


Figure 1: User Flow Diagram of the Domestic Bot Dashboard Kiosk

2.5 Accessibility Plan

Domestic Bot Dashboard is designed to be usable by the widest practical range of museum visitors. Larger Text mode enlarges typography and touch targets for readability from about two metres away, and High Contrast Mode switches to a pure black background with vivid outlines for low-vision visitors. Voice Audio Guidance provides simulated spoken explanations on touch for visitors who prefer or need audio feedback, and Kiosk Reach Mode lowers the active touch zones so seated and younger visitors can comfortably reach every control. The interface supports multiple languages selectable from the header at any time. An always-visible Emergency Stop button, reachable from every screen, immediately opens a confirmation screen that freezes the simulated robot for safety. Together, these controls follow ADA-aligned and WCAG-inspired accessibility conventions while keeping the exhibit's touch interactions simple enough for unattended kiosk use.

Chapter 3

Visual Design & Wireframes (UI)

This chapter presents the exhibit's low-fidelity wireframes, the high-fidelity mockups captured directly from the working application, and the design system that keeps the interface visually consistent.

3.1 Low-Fidelity Wireframes

The low-fidelity wireframes were used to define the basic structure and layout of the kiosk application before visual polish was applied. They show the placement of the header, the robot status card, the AI suggestion banner, the Quick Actions grid, and the two-step task configuration flow, helping to organise the interface and validate the overall information hierarchy early in the design process.

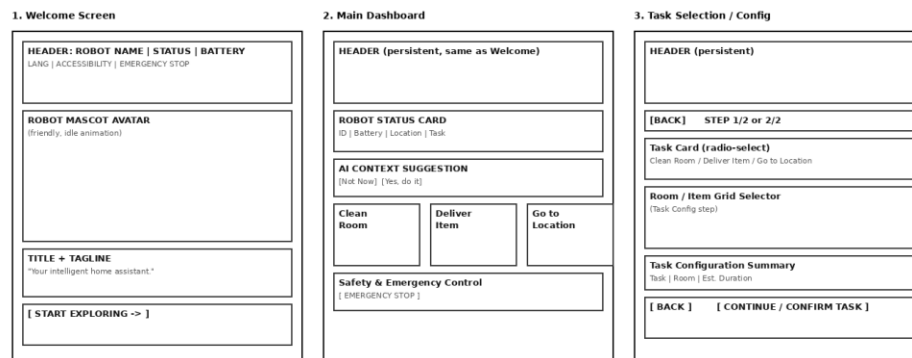


Figure 2: Low-Fidelity Wireframes of the Domestic Bot Dashboard Kiosk

3.2 High-Fidelity Mockups

The high-fidelity mockups below are captured directly from the deployed Domestic Bot Dashboard web application rather than static design files, and show the actual interface layout, colour palette, typography, and live application states. They demonstrate the welcome screen, the main dashboard, and the two-step task selection and configuration flow.

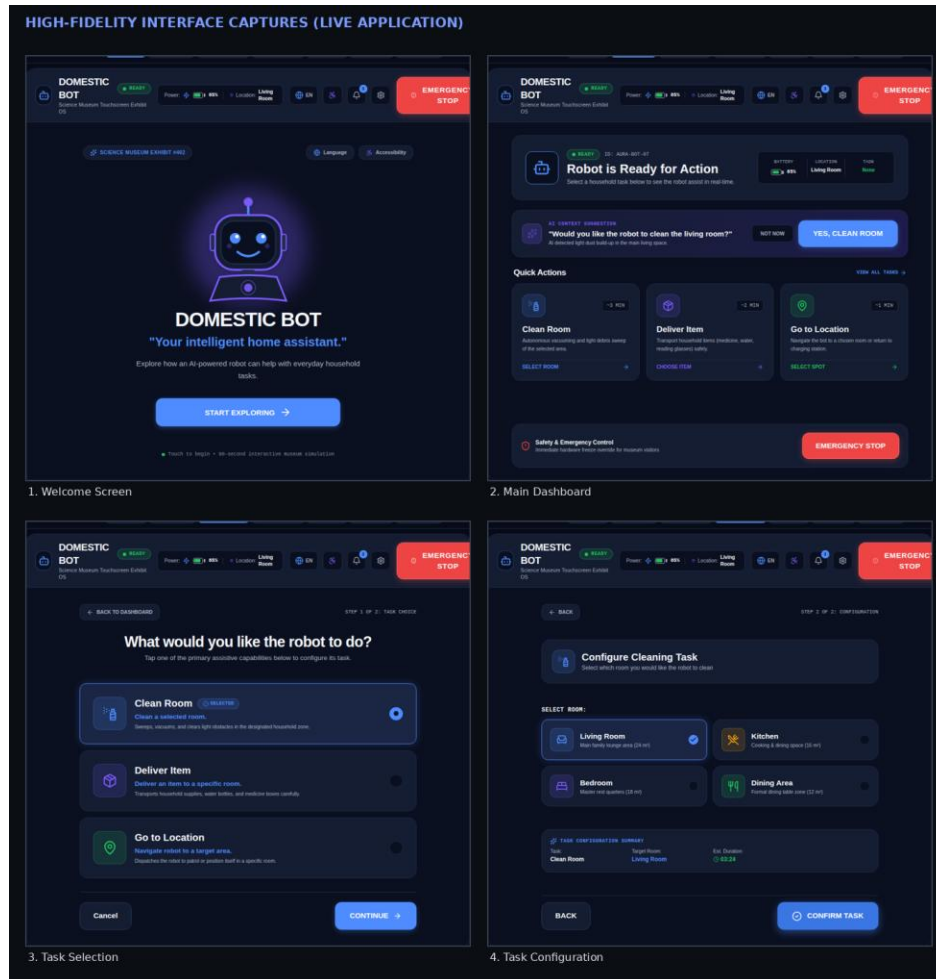


Figure 3: High-Fidelity Mockups — Welcome, Dashboard, and Task Selection Flow (Live Application Captures)

Figure 4 below shows the robot working live, the task-complete feedback screen, the Emergency Stop confirmation, and the Accessibility Controls modal, also captured from the live application.

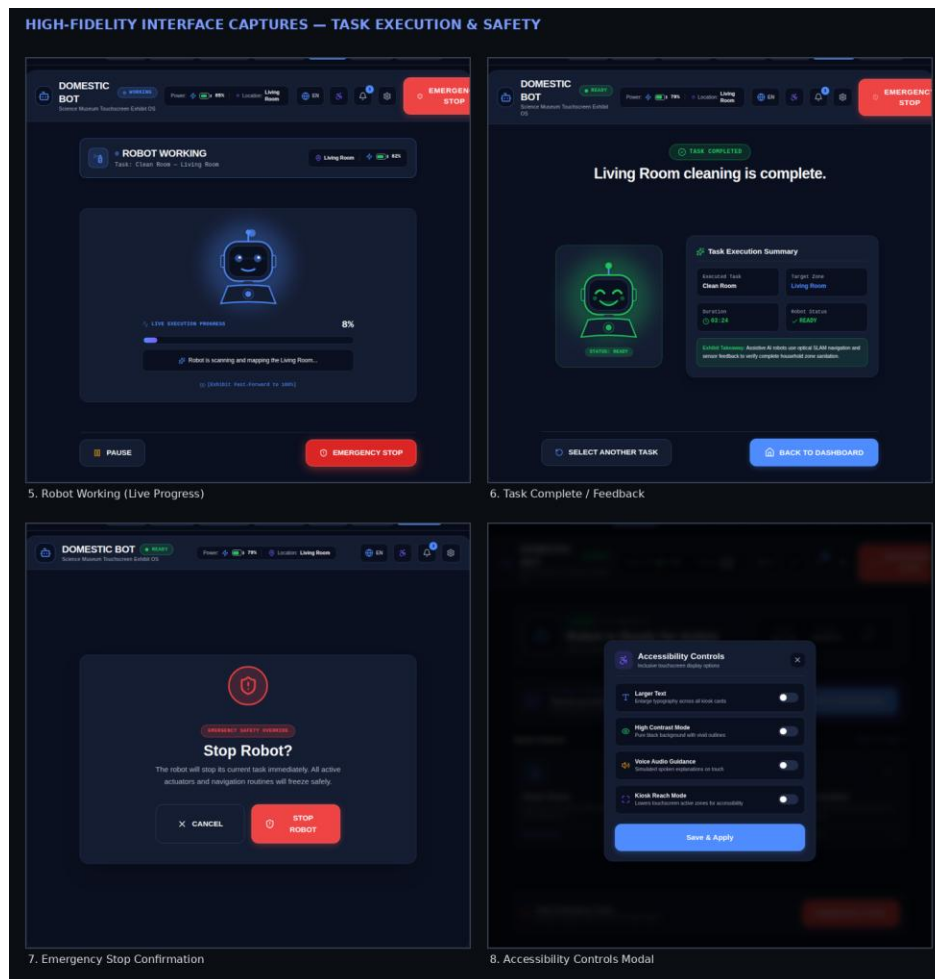


Figure 4: High-Fidelity Mockups — Task Execution, Feedback, and Safety Screens

3.3 System Design / Component Library

The design system defines the visual and interactive elements used consistently throughout the exhibit, keeping the interface coherent across all screens and making it easier for visitors to learn. It includes the colour palette (blue for the primary/active state, purple for secondary actions, green for success, amber for warnings, and red for the emergency state), a clean sans-serif typography scale for headings, labels, and status text, a shared component library (robot status card, battery indicator, AI suggestion banner, quick action cards, task configuration summary), consistent button states, and the ADA-aligned accessibility controls described in Section 2.5.

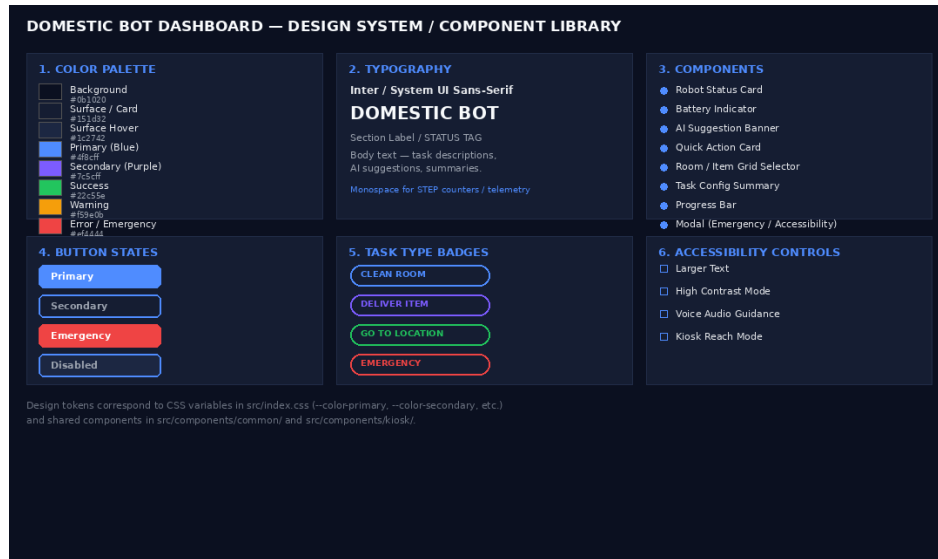


Figure 5: Design System and Component Library (Colour Palette, Typography, Components, Button States, Task Badges, and Accessibility Controls)

Chapter 4

Technical Architecture & Kiosk Specs

This chapter documents the hardware integration and deployment architecture behind the kiosk, the application's internal component architecture, and how visitor input maps onto system responses.

4.1 Hardware Integration Map

Domestic Bot Dashboard was designed for deployment on a 55-inch vertical touchscreen museum kiosk, but requires no specialised hardware beyond a modern touchscreen and a Chromium-based browser: the entire application is a static, client-side bundle. Table 4 summarises the target hardware and software specification, and Figure 6 illustrates how visitor touch input flows through the browser to the application bundle, mirroring the interface that a real robot controller would occupy in a future hardware deployment.

Table 4: Kiosk Hardware & Software Specification

Component	Specification
Display	55-inch vertical touchscreen kiosk display
Input	Capacitive multi-touch
Application	React 19 + TypeScript, bundled with Vite 6
Styling	Tailwind CSS 4
Animation	Motion (Framer Motion successor library)
AI Layer (optional)	Google Gemini API for natural-language task suggestions
Browser	Chromium-based browser in kiosk mode
Network	None required for core kiosk flow (static bundle); optional network only for the AI suggestion feature

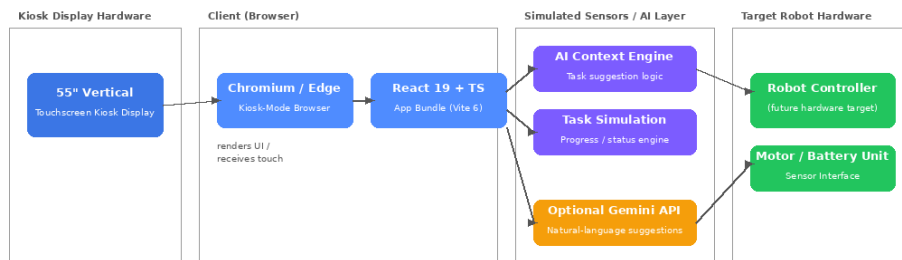


Figure: Hardware Integration Map — touch input flows from the display through the browser-based application to the AI/task-simulation layer; the current build simulates robot execution in-browser, ready for a future physical robot controller.

Figure 6: Hardware Integration Map of the Domestic Bot Dashboard Kiosk

Internally, the application follows a single-source-of-truth component architecture: a root App.tsx component owns the current screen, robot state, task configuration, accessibility settings, and selected language, and passes this state down to stateless screen and presentation components, as shown in Figure 7.

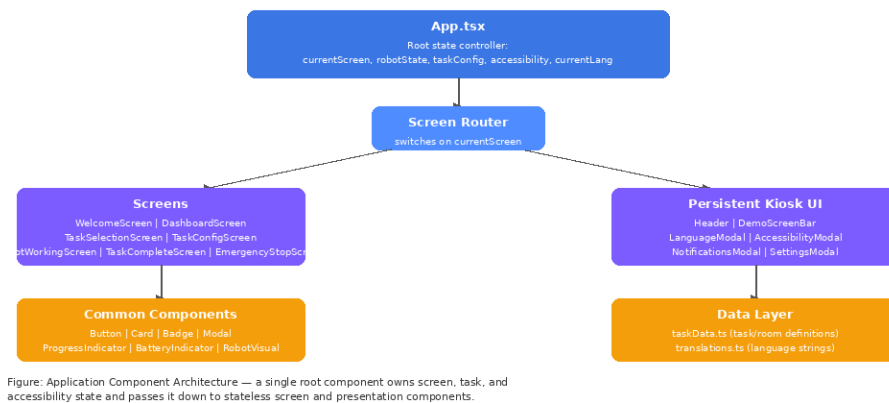


Figure 7: Application Component Architecture

4.2 Sensor & Controller Mapping

The current prototype uses the touchscreen as its sole input method for the core kiosk flow; task execution and robot status are realistically simulated in-browser rather than read from physical sensors, so the interface can be fully demonstrated without robot hardware. Table 5 maps each visitor-facing control to the corresponding system response.

Table 5: Sensor & Controller Mapping

Input / Control	Visitor Action	System Response
Touch / Tap	Tap “Start Exploring” on the Welcome screen	Opens the Main Dashboard
Touch / Tap	Tap a Quick Action or “View All Tasks”	Opens Task Selection
Touch / Tap	Select a task type, tap Continue	Opens Task Configuration for that task
Touch / Tap	Select a room / item, tap Confirm Task	Dispatches the robot; opens Robot Working screen
Touch / Tap	Tap Pause / Resume or Fast-Forward	Pauses, resumes, or accelerates the task simulation
Touch / Tap	Tap Emergency Stop (header or dashboard)	Opens the Emergency Stop confirmation screen
Touch / Tap	Tap “Stop Robot” on the Emergency screen	Freezes all simulated actuators and returns to the dashboard

Input / Control	Visitor Action	System Response
Touch / Tap	Open Accessibility panel, toggle a setting, Save & Apply	Applies larger text, high contrast, voice guidance, or reach mode

Chapter 5

Conclusion

This chapter summarises the project, discusses its limitations, and outlines directions for future work.

5.1 Summary

Domestic Bot Dashboard was developed as an interactive, touch-based museum kiosk exhibit that lets visitors configure and dispatch a simulated assistive household robot, then watch it complete the task with live progress feedback. The system lets visitors choose from a small set of realistic domestic tasks, configure each one in a couple of taps, and receive a clear execution summary once the task completes. The project also focused on multilingual support, ADA-aligned accessibility controls, and an always-reachable Emergency Stop, so that visitors of different ages, backgrounds, and abilities can interact with the exhibit comfortably and safely. Overall, the project demonstrates how UI/UX design, interactive simulation, and modern front-end web technologies can be combined to make an otherwise invisible process — autonomous robot task planning and execution — engaging and easy to understand within a short museum visit.

5.2 Limitation

- The current version is a purely software-based simulation; it does not use a real robot controller or physical sensors, and task execution timing is approximated rather than driven by an actual robotics stack.
- The header's Emergency Stop button has a minor visual overflow at the default kiosk frame width, where its right edge is slightly clipped; this is a cosmetic layout issue that does not affect functionality.
- The interface has not yet been evaluated through a formal usability study with real museum visitors or a structured accessibility audit against a specific WCAG conformance level.

5.3 Future Work

- Connect the dashboard to a real robot controller and physical sensor suite for an on-site hardware deployment.
- Fix the header's minor button-overflow issue and re-verify layout across all supported screen widths.
- Conduct a structured usability study with museum visitors, including visitors using assistive technology, to validate and refine the accessibility implementation.
- Add on-device, non-identifying analytics to help museum staff understand which tasks and languages are most used.

References

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